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Thermodynamics of electrolyte solutions near charged surfaces: Constant surface charge vs constant surface potential

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ABSTRACT

Electric double layers are ubiquitous in science and engineering and are of current interest, owing to their applications in the stabilization of colloidal suspensions and as supercapacitors. While the structure and properties of electric double layers in electrolyte solutions near a charged surface are well characterized, there are subtleties in calculating thermodynamic properties from the free energy of a system with charged surfaces. These subtleties arise from the difference in the free energy between systems with constant surface charge and constant surface potential. In this work, we present a systematic, pedagogical framework to properly account for the different specifications on charged bodies in electrolyte solutions. Our approach is fully variational—that is, all free energies, boundary conditions, relevant electrostatic equations, and thermodynamic quantities are systematically derived using variational principles of thermodynamics. We illustrate our approach by considering a simple electrolyte solution between two charged surfaces using the Poisson–Boltzmann theory. Our results highlight the importance of using the proper thermodynamic potential and provide a general framework for calculating thermodynamic properties of electrolyte solutions near charged surfaces. Specifically, we present the calculation of the pressure and the surface tension between two charged surfaces for different boundary conditions, including mixed boundary conditions.

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I. INTRODUCTION

Electric double layers (EDLs) in electrolyte solutions near a charged surface have a rich history, owing to their relevance in several fields of science and engineering.^{1–13} Copious theories to describe EDL structures and forces have been developed, starting with the original work by Helmholtz¹⁴ in 1879, followed by revisions from Gouy¹⁵ and Chapman¹⁶ to account for ion mobility in a solvent. Later, Stern¹⁷ united the Helmholtz and Gouy and Chapman models to account for both bound ions at the electrode interface (the Stern layer) and the diffuse electric double layer. This Gouy–Chapman–Stern model has been used extensively in the study of EDLs. For strongly charged surfaces, EDLs can exhibit anomalous behaviors such as charge inversion and like-charge attraction.¹⁸ Such behaviors occur for low molecular weight electrolytes in the strong-coupling limit where electrostatic correlations dominate, or for polyelectrolytes even well below the strong-coupling limit.¹ Other electrostatic correlation effects include image charge,^{19–22} where ions

are either attracted or repelled from a surface depending on the nature of the dielectric discontinuity. EDLs remain a subject of great interest owing to their importance in applications such as energy storage in supercapacitors²³ and stabilization of colloidal suspensions.^{2,24–27}

Most theoretical formulations rely on constructing a free energy of the system. Various thermodynamic properties can be calculated from derivatives of this free energy. One notable property in EDLs is the pressure (or force) between charged surfaces. For example, in colloidal systems, according to Derjaguin–Landau–Verwey–Overbeek (DLVO) theory,²⁴ the interplay between EDL repulsion and van der Waals interactions determines the stability of a colloidal suspension. The pressure can be calculated by differentiating a free energy for the system; however, the relevant free energy at equilibrium is different for surfaces with constant surface charge vs constant surface potential. Consequently, the proper free energy must be used to correctly obtain the pressure and other thermodynamic properties.

The difference between the free energies for constant surface charge and surface potential conditions is the energy to charge/discharge bodies at constant surface potential. As other thermodynamic variables vary, energy is required to supply or remove charge from the surfaces to maintain constant surface potential. This was recognized and rigorously accounted for over 50 years ago in the works of Verwey and Overbeek,²⁸ Landau *et al.*,²⁹ and Feynman³⁰ for isolated charged bodies. Likewise, many authors over the last several decades extended this to charged bodies in the presence of electrolyte solutions in a rigorous, consistent manner, through the Legendre transform or a Lagrange multiplier.^{31–39} In recent decades, however, the difference between these free energies has often not been explicitly discussed.^{40–45} As we will explore, the difference in thermodynamic potential amounts to whether surface terms are included in the free energy. Of central importance, some authors do not include surface terms in their free energies and instead absorb these terms into boundary conditions. Other authors explicitly include surface terms in the free energy. Presently, there is no accepted convention on whether surface terms should be included. As we will show later, omitting the surface terms amounts to using a Legendre transformed free energy, relevant only for specified surface potential. While not including surface terms is valid and correct, without an explicit explanation, the role of surface terms in the free energy can cause confusion for the readers.

Gupta *et al.* recently published a study that proposed a framework to treat the thermodynamics of EDLs near charged bodies.⁴⁶ A central point in their work was invoking a term they called U_{charge} to be added to the internal energy for the system, whose form depends on different specifications of the boundary condition: $U_{\text{charge}} = 0$ for constant surface charge and $\int_B \sum_j q_j \psi_j d^2\mathbf{r}$ for constant surface potential, where ψ and q are, respectively, the surface potential and charge in their notation. They applied their framework to both a mean-field model of an electrolyte in the Poisson–Boltzmann approximation and an electrolyte model with strong electrostatic correlations described by a modified Gauss’s law. A motivation for their work is their view that the U_{charge} term was overlooked in several previously published studies^{47–56} (Refs. 30–32, 36, and 44–49 in Ref. 46). Their approach yields correct equations for properties such as the ion-density profile and pressure between charged surfaces; however, some aspects of the thermodynamics presented in Ref. 46 could be derived in a more pedagogical and thermodynamically consistent manner. For example, allowing internal energy to have different forms (and values) depending on whether surface charge or surface potential is specified contradicts the fact that the internal energy is a state function whose value is uniquely specified for a given state no matter what state variable is used to specify that state. Instead, under conditions of specified surface potential, the relevant energy is an enthalpy-like new energy given by the Legendre transform $H = U - \int_B \sum_j q_j \psi_j d^2\mathbf{r}$, where the last term is the work associated with charging/discharging the externally charged body at constant potential; the different specifications of the boundary condition correspond to different thermodynamic ensembles. The constant surface charge and constant surface potential boundary conditions are associated with different thermodynamic potentials (free energies) that are related to each other through the Legendre transform. If one does not use the free energy to calculate thermodynamic quantities, the issue of the Legendre transform does not arise,

as is the case in Refs. 47–56. Furthermore, by writing the differential form of U_{charge} as $\delta U_{\text{charge}} = \int_B \sum_j \psi_j \delta q_j d^2\mathbf{r}$ [Eq. (4) of Ref. 46] instead of the total differential $\delta U_{\text{charge}} = \int_B \sum_j (\psi_j \delta q_j + q_j \delta \psi_j) d^2\mathbf{r}$, one would not be able to produce the correct differential form of the Legendre transformed thermodynamic potential, from which many useful relations (e.g., Maxwell relations) follow.

In this work, we seek to present a concise, systematic, and pedagogical framework to properly account for different specified conditions on charged bodies in electrolyte solutions. Starting from a system with specified surface charge, we construct the appropriate free energy and then perform a Legendre transform to obtain a new free energy for a system with specified surface potential. All governing equations, including the Poisson–Boltzmann equation (PBE), boundary conditions, interplate pressure, etc., are systematically derived from the appropriate free energy through variational conditions and by taking the appropriate thermodynamic derivatives. We note that the work by Reiner and Radke³¹ also uses a variational approach to derive Poisson–Boltzmann and other governing equations and does so rigorously. However, our work seeks to serve as a pedagogical introduction to the most relevant information regarding the different specifications of the charged surfaces, whereas the work of Reiner and Radke³¹ is more fitting for the advanced reader. The fact that confusions still persist despite the existence of their work justifies the need for further clarification. Additionally, our analysis shows that the boundary conditions are a natural consequence of variation of the relevant free energy and emphasizes that surface potential and surface charge are not only boundary conditions but are also important thermodynamic variables for charged systems.

The rest of this article is organized as follows: First, we construct the variational grand free energy, then systematically derive the ion densities and the Poisson–Boltzmann equation for an electrolyte solution in the presence of a general external charge distribution. We then discuss the subtleties of fixed surface charge and fixed surface potential boundary conditions for parallel plates in an electrolyte solution and relate the relevant thermodynamic potentials for each case using the Legendre transform. We further show that the U_{charge} term introduced by Gupta *et al.* naturally arises from the Legendre transform. Additionally, we verify the thermodynamic consistency of our approach by showing that constant surface potential and constant surface charge boundary conditions are a natural consequence of variation of the relevant thermodynamic potential. To emphasize the importance of properly treating the free energy, we discuss two thermodynamic properties—pressure between the surfaces and surface tension. We show that using the incorrect thermodynamic potential leads to a qualitatively incorrect pressure between the plates at constant surface potential. For the surface tension, we show that it is associated with a natural thermodynamic potential, regardless of the boundary condition. Our work concludes with an example of mixed boundary conditions—one plate at constant surface charge and the other at constant surface potential—to demonstrate the simplicity of treating the energetics with such specifications using our approach. We show that mixed boundary conditions predict “unlike-charge repulsion” even at the mean-field Poisson–Boltzmann level, which would not have been captured without using the proper free energy. We emphasize that although variational approaches have been previously used³¹ and the Legendre transform has been implemented in the literature,^{31–39}

here, we present them in a unified, pedagogical framework to clarify the energetics of electrolyte solutions in the presence of charged bodies.

II. VARIATIONAL FREE ENERGY

We begin our pedagogical framework by constructing the variational grand free energy for an electrolyte solution near a general external charge distribution, from which we will later show that the Poisson–Boltzmann equation (PBE) and boundary conditions naturally follow by variation of the grand free energy.^{31,57} We start with a general free energy functional and later take the Poisson–Boltzmann limit; however, taking this limit does not compromise the generality of the approach.

Consider a general system of an external charge distribution with charge density ρ_{ex} submerged in a bath of a symmetric (ion valencies $Z_+ = Z_- = Z$), binary electrolyte solution. The bulk ion concentration is c_B . Since our focus is on developing a framework, we will take the ion valencies to be $Z = 1$ for simplicity. The solvating medium is assumed to be a dielectric continuum with uniform electric permittivity ϵ . The Helmholtz free energy of our system is written as a sum of ideal, excess, and mean-field electrostatic contributions,

$$F = F_{\text{id}} + F_{\text{ele}} + F_{\text{ex}}. \quad (1)$$

The ideal portion is given by the free energy of an inhomogeneous mixture of an ideal gas of ions,

$$\beta F_{\text{id}} = \int d\mathbf{r} c_+(\mathbf{r}) \{ \ln [c_+(\mathbf{r}) v_0^+] - 1 \} + \int d\mathbf{r} c_-(\mathbf{r}) \{ \ln [c_-(\mathbf{r}) v_0^-] - 1 \}, \quad (2)$$

where $\beta = 1/kT$, k is the Boltzmann constant, $c_{\pm}$ are the number densities of the ions, and $v_0^{\pm}$ are the characteristic volumes of the ions. We note that some authors use the thermal wavelength cubed as the volume scale.⁵⁸ The choice of the volume scale is inconsequential—it merely results in a concentration-independent shift in the chemical potential.

The excess term is a general contribution arising from electrostatic correlations and other non-electrostatic interactions between charged particles and depends only on the ion number densities $F_{\text{ex}} = F_{\text{ex}}[c_+, c_-]$. The mean-field electrostatic contribution is simply the Coulomb energy of the system,

$$\beta F_{\text{ele}} = \frac{\beta e^2}{2} \int d\mathbf{r} \int d\mathbf{r}' \rho(\mathbf{r}) C(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}'), \quad (3a)$$

$$C(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi\epsilon|\mathbf{r} - \mathbf{r}'|}, \quad (3b)$$

where ρ is the net mean charge density in the system,

$$e\rho = e(\rho_{\text{ex}} + c_+ - c_-), \quad (4)$$

with e being the elementary charge and ρ_{ex} being the charge density of the external charged body, which includes any surface charge terms. Here, ρ and ρ_{ex} are in units of length^{-3} and charge units have been explicitly written as the elementary charge e . As discussed earlier, several authors^{40,41,43–45} do not include surface charge

terms. This convention amounts to using a Legendre transformed free energy corresponding to specified surface potential conditions, which we will discuss later. Technically, both conventions are valid; however, for clarity and consistency, we believe it is best practice to directly include surface terms in the free energy and Legendre transform thereafter.

Noting an identity for the quadratic form in Eq. (3a), the quadratic interactions can be decoupled at the expense of a new coupling to the variable ψ ,⁵⁹ the mean-electrostatic potential. Applying this identity and integrating by parts, the electrostatic contribution can be written as

$$\beta F_{\text{ele}} = \beta \int d\mathbf{r} \left\{ e\rho(\mathbf{r})\psi(\mathbf{r}) - \frac{\epsilon}{2} [\nabla\psi(\mathbf{r})]^2 \right\}. \quad (5)$$

We note that, in order to recover Eq. (3), ψ satisfies the following variational condition:

$$\frac{\delta\beta F_{\text{ele}}}{\delta\psi} = 0. \quad (6)$$

Upon defining the Bjerrum length $l_B = e^2/4\pi\epsilon kT$ and redefining $\psi = \beta e\psi$ for notational simplicity, we obtain the full variational Helmholtz free energy,

$$\beta F = \int d\mathbf{r} \left\{ (\rho_{\text{ex}} + c_+ - c_-)\psi - \frac{1}{8\pi l_B} (\nabla\psi)^2 + c_+ [\ln(c_+ v_0^+) - 1] + c_- [\ln(c_- v_0^-) - 1] \right\} + \beta F_{\text{ex}}[c_+, c_-]. \quad (7)$$

Since our system is submerged in an electrolyte solution bath, the relevant thermodynamic potential at equilibrium is the grand free energy. The grand free energy is obtained from a Legendre transform,

$$W = F - \int d\mathbf{r} (\mu_+ c_+ + \mu_- c_-). \quad (8)$$

Thus,

$$\beta W = \int d\mathbf{r} \left\{ (\rho_{\text{ex}} + c_+ - c_-)\psi - \frac{1}{8\pi l_B} (\nabla\psi)^2 + c_+ [\ln(c_+ v_0^+) - 1] + c_- [\ln(c_- v_0^-) - 1] - (\beta\mu_+ c_+ + \beta\mu_- c_-) \right\} + \beta F_{\text{ex}}[c_+, c_-]. \quad (9)$$

We note that W also satisfies the variational condition in ψ ,

$$\frac{\delta\beta W}{\delta\psi} = 0. \quad (10)$$

Additionally, by the variational principle of thermodynamic potentials,⁶⁰ equilibrium is obtained from the variational condition of W with respect to the unconstrained internal variables $c_{\pm}$,

$$\frac{\delta\beta W}{\delta c_{\pm}} = 0. \quad (11)$$

The variational conditions in ψ and $c_{\pm}$ yield

$$\nabla^2 \psi = -4\pi l_B (\rho_{\text{ex}} + c_+ - c_-), \quad (12)$$

$$c_{\pm} = \frac{e^{\beta\mu_{\pm}}}{v_0^{\pm}} e^{\mp\psi} \exp\left[-\frac{\delta\beta F_{\text{ex}}}{\delta c_{\pm}}\right]. \quad (13)$$

Taking $F_{\text{ex}} = 0$ recovers the Poisson–Boltzmann limit $c_{\pm} = \frac{e^{\beta\mu_{\pm}}}{v_0^{\pm}} e^{\mp\psi}$. Setting the reference potential such that $\psi = 0$ in the bulk solution, we obtain

$$c_{\pm} = c_B e^{\mp\psi}. \quad (14)$$

Finally, after substituting Eq. (14) into (12), we have

$$\nabla^2\psi = -4\pi l_B [\rho_{\text{ex}} - 2c_B \sinh(\psi)], \quad (15)$$

which is the Poisson–Boltzmann equation. The grand free energy becomes

$$\beta W = \int d\mathbf{r} \left[-\frac{1}{8\pi l_B} (\nabla\psi)^2 + \rho_{\text{ex}}\psi - 2c_B \cosh(\psi) \right]. \quad (16)$$

III. SPECIFIED SURFACE CHARGE

We now examine the subtleties between fixed surface charge and surface potential conditions, beginning with the case of fixed surface charge. Consider a system composed of two parallel plates submerged in an electrolyte solution bath depicted in Fig. 1. The solution contains a symmetric, binary, monovalent electrolyte with bulk ion concentration c_B . The left and right plates have specified surface charges Q_1 and Q_2 , respectively, and corresponding *unknown* surface potentials $\mathcal{V}_1$ and $\mathcal{V}_2$. The plates have surface area A and are separated by distance L . The surface charge densities are then $\sigma_1 = Q_1/A$ and $\sigma_2 = Q_2/A$. Making use of the equations derived in the last section, the Poisson–Boltzmann equation (PBE) and the resulting grand free energy for our system become

$$\frac{d^2\psi}{dz^2} = -4\pi l_B [\sigma_1 \delta(z) + \sigma_2 \delta(z-L) - 2c_B \sinh(\psi)], \quad (17)$$

$$\beta W = A \int_0^L dz \left[-\frac{1}{8\pi l_B} \left(\frac{d\psi}{dz} \right)^2 - 2c_B \cosh(\psi) \right] + \psi(0)Q_1 + \psi(L)Q_2, \quad (18)$$

where we have invoked $\rho_{\text{ex}} = \sigma_1 \delta(z) + \sigma_2 \delta(z-L)$ to explicitly include the surface charge for parallel plates and $\delta(z)$ is the Dirac delta function.

Since we specify the plates in Fig. 1 to have fixed surface charge, namely, Q_1 on the left plate and Q_2 on the right, the system is described by a $T, V, A, \mu_{\pm}, Q_1, Q_2$ ensemble. We emphasize that Q_1 and Q_2 are treated as thermodynamic variables that define the system. Accordingly, the relevant free energy at equilibrium is the grand free energy, $W(T, V, A, \mu_{\pm}, Q_1, Q_2)$. The differential form of W is

$$dW = -SdT - PdV + \gamma dA - N_+ d\mu_+ - N_- d\mu_- + e\mathcal{V}_1 dQ_1 + e\mathcal{V}_2 dQ_2, \quad (19)$$

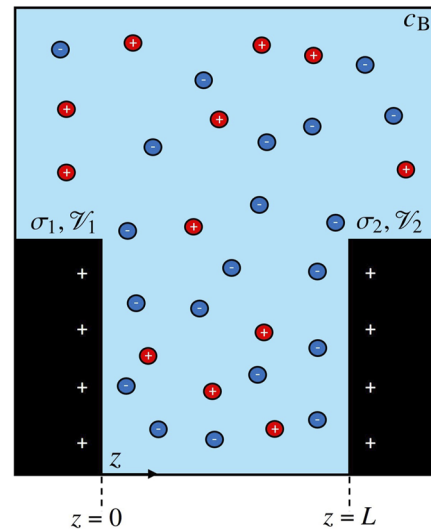


FIG. 1. A symmetric binary electrolyte solution containing two fully submerged parallel plates in contact with an infinitely large reservoir. The bulk solution has ion concentration c_B , and the ions have valency $Z = 1$. The left and right plates have surface charge density and surface potential $\sigma_1, \mathcal{V}_1$ and $\sigma_2, \mathcal{V}_2$, respectively, and are located at $z = 0$ and $z = L$. Here, the surface charge densities and surface potentials are in units of length^{-2} and energy/charge, respectively. The plates have area A and are separated by distance L . The system volume is $V = AL$.

where S is the system entropy, P is the pressure in the film (the region of electrolyte solution between the parallel plates), γ is the surface tension, and N_+ and N_- are the number of positive and negative ions given by $N_{\pm} = \int_V d\mathbf{r} c_{\pm}$. Because the solvent is treated implicitly, P is actually the osmotic pressure while γ is the excess surface tension with respect to the surface tension between the solid surface and the pure solvent. Henceforth, we will use the simpler terms pressure and surface tension, respectively, to refer to these quantities. The $\mathcal{V}dQ$ terms correspond to the electrostatic work required to add dQ charges to the surface at potential $\mathcal{V}$. Note that we have explicitly included the elementary charge, e , to remain consistent with our definition of the charge density in Sec. II. For conciseness, the surface tension refers to the surface tension of the two surfaces ($\gamma = \gamma_1 + \gamma_2$) since we do not separately consider each solid–liquid interface. Finally, we note that a differential change in volume is clearly coupled to the differential change in area ($dV = AdL + LdA$) for the confined system. We will revisit the pressure and surface tension later on.

As discussed earlier, W satisfies the variational conditions Eqs. (10) and (11), with respect to ψ and the internal unconstrained variables $c_{\pm}$, and these conditions yield the PBE given by Eq. (17). Thus, the PBE determines ψ and $c_{\pm}$ and is a natural consequence of variation of W . Moreover, the PBE naturally admits fixed surface charge boundary conditions, obtained by integrating the PBE from $z = 0^-$ to $z = 0^+$ for the left plate and $z = L^-$ to $z = L^+$ for the right plate,

$$\left. \frac{d\psi}{dz} \right|_{z=0^+} - \left. \frac{d\psi}{dz} \right|_{z=0^-} = -4\pi l_B \sigma_1, \quad (20)$$

$$\left. \frac{d\psi}{dz} \right|_{z=L^+} - \left. \frac{d\psi}{dz} \right|_{z=L^-} = -4\pi l_B \sigma_2, \quad (21)$$

where the $2c_B \cosh(\psi)$ term dropped out by continuity of ψ . Noting the electric field vanishes inside each plate by Gauss's law and charge neutrality, we obtain fixed surface charge boundary conditions,

$$\sigma_1 = -\frac{1}{4\pi l_B} \left. \frac{d\psi}{dz} \right|_{z=0}, \quad (22)$$

$$\sigma_2 = \frac{1}{4\pi l_B} \left. \frac{d\psi}{dz} \right|_{z=L}. \quad (23)$$

Furthermore, we take advantage of the differential form of W in Eq. (19) to obtain expressions for the thermodynamic conjugates of the surface charges, the surface potentials $\mathcal{V}_1$ and $\mathcal{V}_2$,

$$e\mathcal{V}_1 = \left(\frac{\partial W}{\partial Q_1} \right)_{T,V,A,\mu_{\pm},Q_2}, \quad e\mathcal{V}_2 = \left(\frac{\partial W}{\partial Q_2} \right)_{T,V,A,\mu_{\pm},Q_1}. \quad (24)$$

To perform these derivatives, we invoke the functional chain rule,

$$\left(\frac{\partial W}{\partial Q_1} \right)_{T,V,A,\mu_{\pm},Q_2} = \left(\frac{\partial W}{\partial Q_1} \right)_{T,V,A,\mu_{\pm},Q_2,\psi,c_{\pm}} + \int_0^L \frac{\delta W}{\delta \psi} \frac{\partial \psi}{\partial Q_1} dz + \int_0^L \frac{\delta W}{\delta c_{\pm}} \frac{\partial c_{\pm}}{\partial Q_1} dz, \quad (25)$$

and similarly for the Q_2 derivative. In principle, ψ and $c_{\pm}$ are functions of Q_1 and Q_2 . However, we see the importance of the variational conditions Eqs. (10) and (11), which remove the last two terms of Eq. (25). Accordingly, the Q_i derivatives can be taken without regard to the dependence of ψ and $c_{\pm}$ on Q_i . Evaluating these derivatives in Eq. (24) yields

$$\beta e\mathcal{V}_1 = \psi(0; Q_1, Q_2), \quad \beta e\mathcal{V}_2 = \psi(L; Q_1, Q_2). \quad (26)$$

The function dependence on other state variables ($T, V, A, \mu_{\pm}$) is omitted for notational brevity, and this convention will be adopted for the remainder of the article. Thus, variation of W in Q_1 and Q_2 reveals that the conjugate variables to Q_1 and Q_2 , $\mathcal{V}_1$ and $\mathcal{V}_2$, respectively, are naturally determined from the electrostatic potential evaluated at the surface of each plate. Therefore, under fixed surface charge conditions, variation of the proper free energy, W , naturally admits fixed surface charge boundary conditions and relationships for the conjugate surface potentials.

IV. SPECIFIED SURFACE POTENTIAL

If the plates have specified surface potentials, namely, $\mathcal{V}_1$ and $\mathcal{V}_2$, then the system is described by a $T, V, A, \mu_{\pm}, \mathcal{V}_1, \mathcal{V}_2$ ensemble, and the relevant free energy at equilibrium is now different from W . The new, relevant thermodynamic potential, denoted Y , is given by the Legendre transform of the grand free energy with

respect to the surface charge of each plate fixed at constant surface potential,^{29,32–38}

$$Y[\psi; \mathcal{V}_1, \mathcal{V}_2] = W[\psi; Q_1, Q_2] - e\mathcal{V}_1 Q_1 - e\mathcal{V}_2 Q_2, \quad (27)$$

and, using Eq. (19), the differential of this potential is

$$dY = -SdT - PdV + \gamma dA - N_+ d\mu_+ - N_- d\mu_- - eQ_1 d\mathcal{V}_1 - eQ_2 d\mathcal{V}_2, \quad (28)$$

where Q_1 and Q_2 are now unconstrained internal variables of Y to be determined by minimization of Y at specified surface potentials $\mathcal{V}_1$ and $\mathcal{V}_2$. This new potential Y can be interpreted as a “grand-like” free energy, where the surface potentials are fixed and the surface charges can fluctuate. Physically, the Legendre transform subtracts the electrostatic work associated with charging/discharging the plates at constant surface potential. Thus, the change in Y bounds the non-electrostatic work for a constant surface potential process taking place at isothermal conditions with fixed reservoir chemical potentials. Furthermore, note that $eQ_1 \mathcal{V}_1 + eQ_2 \mathcal{V}_2$ is equivalent to U_{charge} from Ref. 46 for the case of parallel plates. Thus, our analysis provides the thermodynamic foundation for this term that was introduced phenomenologically in Ref. 46. Finally, our analysis shows that U_{charge} should only be included for thermodynamic potentials that correspond to specified surface potential.

To verify the thermodynamic consistency of free energy Y , we note that Y also satisfies the variational conditions in ψ and $c_{\pm}$,

$$\frac{\delta Y}{\delta \psi} = 0, \quad (29a)$$

$$\frac{\delta Y}{\delta c_{\pm}} = 0. \quad (29b)$$

Applying these conditions to Y returns the Poisson–Boltzmann equation as expected. As mentioned before, Q_1 and Q_2 are now internal unconstrained variables. Therefore, at equilibrium, Y satisfies the variational conditions,

$$\left(\frac{\partial Y}{\partial Q_1} \right)_{T,V,A,\mu_{\pm},\mathcal{V}_1,\mathcal{V}_2;Q_2,\psi,c_{\pm}} = 0, \quad (30)$$

$$\left(\frac{\partial Y}{\partial Q_2} \right)_{T,V,A,\mu_{\pm},\mathcal{V}_1,\mathcal{V}_2;Q_1,\psi,c_{\pm}} = 0. \quad (31)$$

In the subscripts of Eqs. (30) and (31), we use a semicolon to separate system-specifying variables $T, V, A, \mu_{\pm}, \mathcal{V}_1$, and $\mathcal{V}_2$ from the unconstrained variational variables Q_1, Q_2, ψ , and $c_{\pm}$. These derivatives are taken at constant Q_2 and Q_1 , respectively, because they are partial derivatives. Additionally, Eqs. (30) and (31) are taken at constant ψ and $c_{\pm}$ due to variational conditions on ψ and $c_{\pm}$ as discussed in Sec. III. Applying the variational conditions on Q_i to the Legendre transform given in Eq. (27) yields

$$\left(\frac{\partial W}{\partial Q_1} \right)_{T,V,A,\mu_{\pm},\mathcal{V}_1,\mathcal{V}_2;Q_2,\psi,c_{\pm}} = e\mathcal{V}_1, \quad (32)$$

$$\left(\frac{\partial W}{\partial Q_2}\right)_{T,V,A,\mu_{\pm},V_1,V_2,Q_1,\psi,c_{\pm}} = eV_2. \quad (33)$$

Evaluating these derivatives gives

$$\psi(0; Q_1, Q_2) = \beta e V_1, \quad \psi(L; Q_1, Q_2) = \beta e V_2. \quad (34)$$

Therefore, at specified surface potential, constant surface potential boundary conditions are a natural consequence of variation of the proper free energy Y and are given by setting the electrostatic potential at the surface of the plate to be the specified surface potential. These boundary conditions also serve as an implicit equation to determine the unknown surface charges, Q_1 and Q_2 —equations of state of the form $Q_1 = f(V_1, V_2)$ and $Q_2 = f(V_1, V_2)$. We emphasize that both W and Y naturally produce the same $Q_1 = f(V_1, V_2)$ and $Q_2 = f(V_1, V_2)$ equations of state, Eqs. (26) and (34), as expected by thermodynamic consistency. The only subtlety to highlight is that Eq. (26) expresses V_1 and V_2 as functions of Q_1 and Q_2 , and Eq. (34) expresses Q_1 and Q_2 as functions of V_1 and V_2 , but are still equivalent equations of state. This is analogous to the equation of state of a gas, where one can either obtain the pressure by specifying the density in a canonical ensemble, or determine the density by specifying the pressure in an isobaric ensemble. Noting the boundary conditions in Eq. (34), we can now apply the Legendre transform in Eq. (27) to W and obtain an expression for Y ,

$$\beta Y = A \int_0^L dz \left[-\frac{1}{8\pi l_B} \left(\frac{d\psi}{dz} \right)^2 - 2c_B \cosh(\psi) \right]. \quad (35)$$

Compared to W in Eq. (18), Y does not explicitly contain the boundary terms. Therefore, we see that omitting the surface charge terms in W would give Y , amounting to a Legendre transformed free energy.

V. INTERPLATE FORCE

We now provide a brief analytical calculation of the interplate force. The pressure in the film P is related to the pressure of bulk reservoir P_{bulk} by the disjoining pressure $\Pi_D \equiv P - P_{\text{bulk}}$. At constant surface charge, the pressure between the plates is obtained from a derivative of W ,

$$PA = -\left(\frac{\partial W}{\partial L}\right)_{T,A,\mu_{\pm},Q_1,Q_2}. \quad (36)$$

The partial derivative is taken at constant A ; thus, the differential volume is proportional to the differential in the plate separation L . Similarly, at constant surface potential, the pressure is obtained from a derivative of Y ,

$$PA = -\left(\frac{\partial Y}{\partial L}\right)_{T,A,\mu_{\pm},V_1,V_2}. \quad (37)$$

It can be shown that, upon applying Eqs. (36) and (37) to Eqs. (18) and (35), respectively, and making use of the Poisson–Boltzmann equation, Eq. (17), both derivatives result in the general expression

$$\beta P = -\frac{1}{8\pi l_B} \left(\frac{d\psi}{dz} \right)^2 + 2c_B \cosh(\psi), \quad (38)$$

which can be evaluated at any position since the pressure is constant between the plates by mechanical equilibrium. The derivation of Eq. (38) requires careful manipulation of the free energy and is deferred to the Appendix. The first term on the right-hand side of Eq. (38) is the isotropic part of the Maxwell stress tensor,⁶¹ and the second term is the entropic contribution from the ions in the presence of potential ψ . For the symmetric case ($Q_1 = Q_2 = Q$ or $V_1 = V_2 = V$), the gradient of the potential at the midplane is zero so that

$$\beta P = 2c_B \cosh(\psi|_{L/2}). \quad (39)$$

Equation (39) is an equation of state known from the contact value theorem.² As expected, Eqs. (36) and (37) yield equivalent expressions because a system's equilibrium state in a reversible process is independent of boundary conditions. In fact, Eq. (39), together with the V – Q equation of state [either Eq. (26) or (34)], constitutes the P – V or P – Q equation of state. Both are equivalent—one can simply interchange between P as a function of V and P as a function of Q using the V – Q equation of state. Furthermore, specifying Q or specifying V in a reversible process simply corresponds to different paths taken on a V – Q surface. Changing the path taken affects the work done and heat exchanged, but the same pressure equation of state is valid everywhere along any path. This is analogous to an ideal gas, where the ideal gas law is valid regardless of the process (e.g., adiabatic, isothermal). However, we stress that to obtain Eq. (39) from a thermodynamic derivative, it is necessary to use the correct thermodynamic potential for the specified conditions. A straightforward substitution of Eq. (27) into (37) for symmetric plate surface potentials gives the following identity:

$$\begin{aligned} PA &= -\left(\frac{\partial Y}{\partial L}\right)_{T,A,\mu_{\pm},V} = -\left(\frac{\partial(W - 2eVQ)}{\partial L}\right)_{T,A,\mu_{\pm},V} \\ &= -\left(\frac{\partial W}{\partial L}\right)_{T,A,\mu_{\pm},V} + 2eV\left(\frac{\partial Q}{\partial L}\right)_{T,A,\mu_{\pm},V}. \end{aligned} \quad (40)$$

This identity can also be derived directly from Eq. (19) by dividing by dL and applying constant temperature, chemical potentials, and surface potential. From Eq. (40), attempting to obtain the pressure solely by differentiating W at constant surface potential neglects the last term in Eq. (40), which accounts for the charging/discharging of the plates as they move. We will explore the relative importance of the last two terms later.

VI. SURFACE TENSION

We close our discussion of thermodynamic consistency with a brief discussion of the surface tension of an electrolyte solution, an important quantity in studying the wettability of surfaces, especially in electrowetting applications.⁶² The thermodynamic definition of surface tension introduced by Gibbs⁶³ is commonly used to calculate surface tension. The basis of Gibbs's formulation is to divide the inhomogeneous system into homogeneous *bulk* phases separated by a *surface* region. The majority of discussion in the time since has been focused on the application of Gibbs's concept of dividing surfaces to curved surfaces and cases where the system is not macroscopic.^{64–72} Here, we only wish to clarify the role of charged

solid surfaces on the surface tension. To this end, we calculate surface tension in a confined system with charged, planar surfaces. We find that the natural thermodynamic potential to evaluate the surface tension is related to the Legendre transform between constant surface charge and surface potential. From the differential form of W and Y ,

$$\gamma = \left(\frac{\partial W}{\partial A} \right)_{T,V,\mu_{\pm},Q_1,Q_2} = \left(\frac{\partial Y}{\partial A} \right)_{T,V,\psi_{\pm},\nu_1,\nu_2}. \quad (41)$$

Writing $dV = AdL + LdA$ in Eqs. (19) and (28), we also have

$$\begin{aligned} \gamma A &= A \left(\frac{\partial W}{\partial A} \right)_{T,L,\mu_{\pm},Q_1,Q_2} + PV \\ &= A \left(\frac{\partial Y}{\partial A} \right)_{T,L,\psi_{\pm},\nu_1,\nu_2} + PV. \end{aligned} \quad (42)$$

Since at fixed $T, L, \mu_{\pm}, \nu_1, \nu_2$, the potential Y is extensive in area A , the derivative in the second line of Eq. (42) is simply Y/A . Therefore,

$$\gamma A = Y + PV. \quad (43)$$

The last term on the right-hand side of Eq. (43) corresponds to the grand potential of a homogeneous system at film pressure P , i.e., $PV = (\Pi_D + P_{\text{bulk}})V = -W_0 = -Y_0$, where subscript 0 indicates a homogeneous system, and for such a system, $W_0 = Y_0$, since there are no surface charge or potential terms.

Equation (43) can be understood from another perspective. Since W is a first-order, homogeneous equation in its extensive variables (V, A, Q_1, Q_2), we have

$$\begin{aligned} W &= V \left(\frac{\partial W}{\partial V} \right)_{T,A,\mu_{\pm},Q_1,Q_2} + A \left(\frac{\partial W}{\partial A} \right)_{T,V,\mu_{\pm},Q_1,Q_2} \\ &+ Q_1 \left(\frac{\partial W}{\partial Q_1} \right)_{T,V,A,\mu_{\pm},Q_2} + Q_2 \left(\frac{\partial W}{\partial Q_2} \right)_{T,V,A,\mu_{\pm},Q_1}. \end{aligned} \quad (44)$$

The partial derivatives on the right-hand side can be identified from the differential form of W in Eq. (19), leading to the following Euler equation:

$$W - eQ_1\nu_1 - eQ_2\nu_2 = -PV + \gamma A. \quad (45)$$

The left-hand side is exactly the definition of Y from the Legendre transform of W .

The surface tension is commonly evaluated using the integral form of the free energy such that $\gamma A = \Omega - \Omega_{\text{bulk}} = \Omega + P_{\text{bulk}}V$, where Ω is a type of grand free energy. The *bulk* subscript indicates that the interface is in contact with a macroscopic, bulk phase. The confined system has an additional $\Pi_D V$ term due to the difference in the film pressure from the pressure of the bulk reservoir. For a charged system, the common expression for surface tension is only valid when Ω is the “grand-like” free energy Y , not the grand free energy W , regardless of whether the system is characterized by constant surface charge or constant surface potential.

Within PB theory, one can analytically obtain the result for the surface tension using the previous expressions for Y from Eq. (35) and P from Eq. (38) in Eq. (43),

$$\beta\gamma = -\frac{1}{4\pi l_B} \int_0^L dz \left(\frac{d\psi}{dz} \right)^2. \quad (46)$$

We see that, like in the pressure expression Eq. (38), the electric field generates a negative contribution to surface tension, reflecting the effect of the Maxwell stress.⁶¹ The usual definition is identical to the definition using Y when $P = P_{\text{bulk}}$, which strictly corresponds to the limit where $L \rightarrow \infty$. Using the PB equation, one can analytically evaluate this limit to obtain the surface tension when the two surfaces are infinitely separated,

$$\beta\gamma_{\infty} = -\frac{2}{\pi\lambda_D l_B} \left\{ \sinh^2 \left[\frac{\psi(0)}{4} \right] + \sinh^2 \left[\frac{\psi(L)}{4} \right] \right\}, \quad (47)$$

where λ_D is the Debye screening length given by $\lambda_D = 1/\sqrt{8\pi l_B c_B}$. In the expression for γ_{∞} , the separate contributions from each surface are additive. Note that the above expression is valid even when the two surfaces are at different potentials. Both Eqs. (46) and (47) were derived by Reiner and Radke,³¹ who also presented analytical results in the Debye–Hückel limit. In the special case of symmetric plates at infinite separation, the total interfacial tension is twice that for a single surface. Substituting the exact relationship between ψ and σ for a single plate into Eq. (47), we obtain

$$\beta\gamma_{\infty} = \frac{-2\sigma^2 \sqrt{2\pi \frac{l_B}{c_B}}}{1 + \sqrt{\frac{\pi\sigma^2 l_B}{2c_B} + 1}}. \quad (48)$$

The thermodynamic potential Y naturally leads to a useful Maxwell relation involving the surface tension. From the differential form of the free energy in Eq. (28), one has the following Maxwell relation:

$$\left(\frac{\partial \gamma}{\partial e\nu_1} \right)_{T,V,\mu_{\pm},\nu_2} = - \left(\frac{\partial Q_1}{\partial A} \right)_{T,V,\mu_{\pm},\nu_2} = -\sigma_1, \quad (49)$$

with a similar equation in which indices 1 and 2 are switched. The equation above can be considered to be a general form of the Lippmann equation⁷³ for the confined system. In the limit of infinite surface separation or for a single surface, one can write the more common form, which has been noted by several authors,^{74–78}

$$\left(\frac{\partial \gamma}{\partial e\nu} \right)_{T,V,\mu_{\pm}} = - \left(\frac{\partial Q}{\partial A} \right)_{T,V,\mu_{\pm}}. \quad (50)$$

VII. ILLUSTRATIVE EXAMPLES

As we showed above, simple analytical expressions for the pressure and surface tension can be obtained at the Poisson–Boltzmann level. This is not always possible (e.g., complex free energy functionals), and one may need to numerically differentiate or use the integral form of the free energy to obtain the quantities. However, differentiating W under constant surface potential conditions yields an incorrect pressure. This can be seen analytically from the identity given by Eq. (40). Likewise, using $\gamma A = W + PV$ yields the incorrect surface tension, regardless of the boundary condition. We first demonstrate the calculation of the pressure by numerically evaluating the pressure under constant surface charge density and constant surface potential conditions for the system depicted in Fig. 1.

Figure 2(a) shows the disjoining pressure in the electrolyte solution as a function of plate separation for symmetric plates at constant

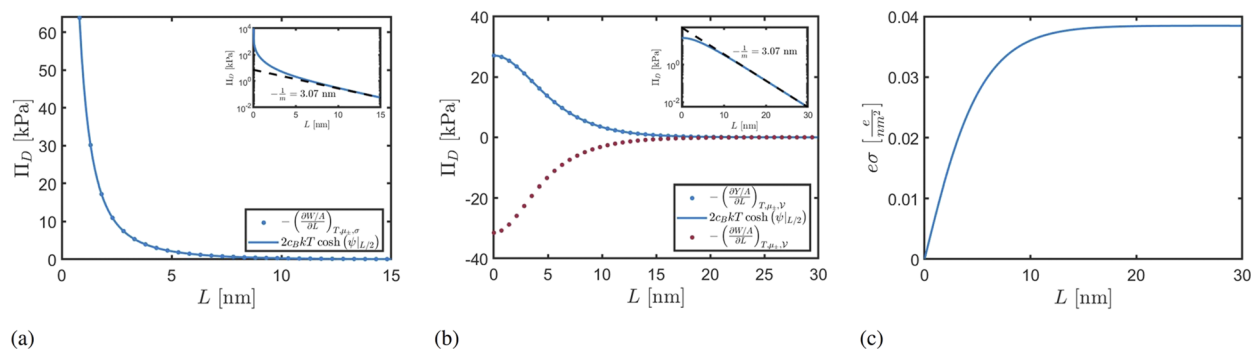


FIG. 2. Disjoining pressure $\Pi_D \equiv P - P_{\text{bulk}}$ and surface charge density for an electrolyte solution at constant surface charge density and constant surface potential. The solution has bulk concentration $c_B = 0.01$ M, temperature $T = 300$ K, and Bjerrum length $l_B = 0.7$ nm. (a) Pressure at constant surface charge density $e\sigma = 0.01$ e/nm², calculated by numerically differentiating Eq. (18) (blue dots) and theoretically (blue line) with Eq. (39). (b) Pressure at constant surface potential $\mathcal{V} = 1$ kT/e, calculated by numerically differentiating Eq. (35) (blue dots), theoretically (blue line) with Eq. (39), and incorrectly evaluated (red dots) by numerically differentiating W , the first term on the rightmost side of Eq. (40). (c) Surface charge density on the plates in (b). Insets: semilog plot of the theoretical pressure [Eq. (39)], where m is the slope of the curve in the linear regime.

surface charge density ($\sigma_1 = \sigma_2 = \sigma$). As the plates come together, the pressure diverges due to the repulsion of the positive charges on the plates. The initial decay with increasing distance is very rapid, stemming from nonlinearities at high surface potential. The rapid decay is followed by a long-range, exponential decay arising from electrostatic screening of the plates due to the electric double layer present at the plate surfaces, with a length scale dictated by the Debye length. This result agrees with other authors.^{36,42}

Figure 2(b) shows the disjoining pressure as a function of plate separation at constant surface potential ($\mathcal{V}_1 = \mathcal{V}_2 = \mathcal{V}$), calculated by numerically differentiating Y given by Eq. (35) (blue dots) and theoretically using Eq. (39) (blue line). The numerically correct and theoretical curves are in agreement and predict repulsion between the plates at all separation distances. Moreover, the repulsion is weaker at close separation for constant surface potential compared to constant surface charge density. At constant surface potential, the plates discharge to satisfy the boundary conditions [see Fig. 2(c)];

however, for fixed surface charge density, the surface charge density remains constant and the surface potential and force diverge at close separation. Another important feature is that the curves saturate at zero separation for the case of constant surface potential. Interestingly enough, this saturation relates to a solvation energy for the plates. The area under the pressure curve is the negative of the free energy of separating the two surfaces at contact—when the surfaces are not exposed to the electrolyte solution—to infinity—when the two surfaces are fully solvated by the solution. Although the decay length of the pressure at constant surface potential appears to be larger than at constant surface charge density, the inset plots of Figs. 2(a) and 2(b) show that both decay lengths are the same and equal to the Debye screening length, $\lambda_D = 3.07$ nm, for the given conditions. Finally, from Fig. 2(b), the red curve [given by the penultimate term of Eq. (40)] predicts attraction between the plates despite both being like-charged—a qualitatively incorrect result. The result is further invalidated by Fig. 2(c), where we see that the

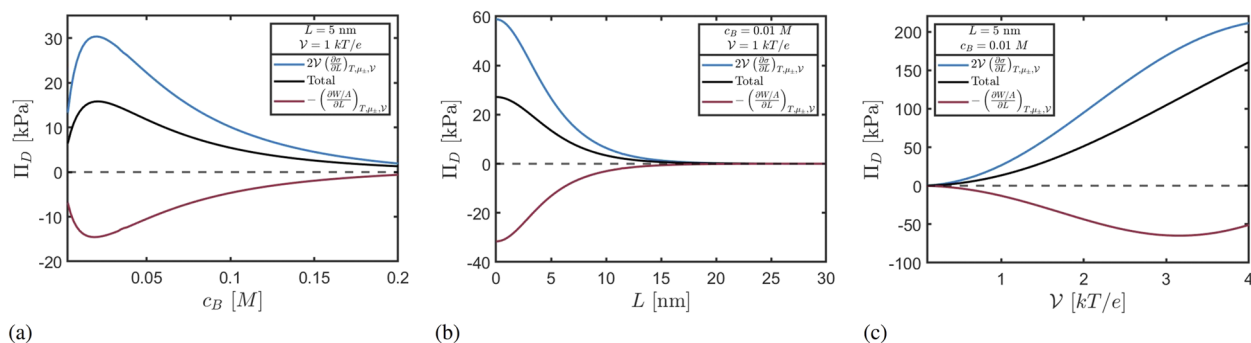


FIG. 3. Relative contributions of the last two terms of Eq. (40) to Π_D as a function of c_B , L , and $\mathcal{V}$. The solution conditions are $T = 300$ K and $l_B = 0.7$ nm. The dashed line shows the zero line for clarity. (a) Pressure contributions vs bulk concentration c_B at $L = 5$ nm and $\mathcal{V} = 1$ kT/e. (b) Pressure contributions vs plate separation L at $c_B = 0.01$ M and $\mathcal{V} = 1$ kT/e. (c) Pressure contributions vs surface potential $\mathcal{V}$ at $c_B = 0.01$ M and $L = 5$ nm.

plates are positively charged at all separation distances and, thus, are expected to always repel.

Now, we explore the relative importance of the last two terms in Eq. (40) for various c_B , L , and $\mathcal{V}$. In Fig. 3, we see that the last two terms of Eq. (40) are nearly equally important for all conditions considered. Moreover, the contribution to the pressure from the charging/discharging of the plates [blue line, the last term of Eq. (40)] is always repulsive, whereas the contribution from the grand free energy [red curve, the penultimate term of Eq. (40)] is always attractive. As an aside, the total pressure in Fig. 3(b) was calculated by adding the individual terms on the right hand side of Eq. (40), $-(\frac{\partial W}{\partial L})_{T,A,\mu_{\pm},\mathcal{V}}$ and $2e\mathcal{V}(\frac{\partial Q}{\partial L})_{T,A,\mu_{\pm},\mathcal{V}}$. On the other hand, the total pressure in Fig. 2(b) was calculated by directly differentiating Y given by Eq. (35). Both curves match, thus proving the equality of Eq. (40) and further verifying the thermodynamic consistency of our approach. Finally, we can perform a final check of thermodynamic consistency using a Maxwell relation. From the differential form of Y given by Eq. (28), we can write the following Maxwell relation for symmetric plates:

$$\left(\frac{\partial P}{\partial e\mathcal{V}}\right)_{T,A,\mu_{\pm}} = 2\left(\frac{\partial Q}{\partial \mathcal{V}}\right)_{T,A,\mu_{\pm}} = 2\left(\frac{\partial \sigma}{\partial L}\right)_{T,\mu_{\pm}}. \quad (51)$$

Evaluating the left and right derivatives from Figs. 2(c) and 3(c), respectively, under the same conditions ($\mathcal{V} = 1 \text{ kT}/e$, $L = 5 \text{ nm}$, and $c_B = 0.01 \text{ M}$) gives $\partial P/\partial \mathcal{V} = 2e\partial \sigma/\partial L = 6.4 \times 10^{-3} \text{ e/nm}^3$, which confirms the thermodynamic consistency.

Turning to the surface tension, we first examine the scaling behavior of the surface tension at infinite plate separation γ_{∞} . The results are shown in Fig. 4, where γ_{∞} is calculated using Eq. (48) as a function of surface charge density for various bulk concentrations. From Fig. 4, we see that the magnitude of the surface tension scales quadratically for low σ , but becomes linear for large σ . The surface tension also loses dependence on the bulk concentration at large σ (we note, however, that this regime may exceed

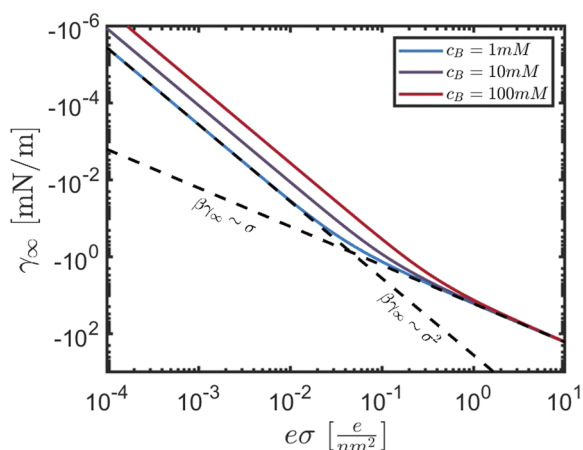


FIG. 4. Surface tension at infinite plate separation of a confined electrolyte solution as a function of surface charge density for various bulk concentrations, calculated from Eq. (48). The solution conditions are $T = 300 \text{ K}$ and $l_B = 0.7 \text{ nm}$.

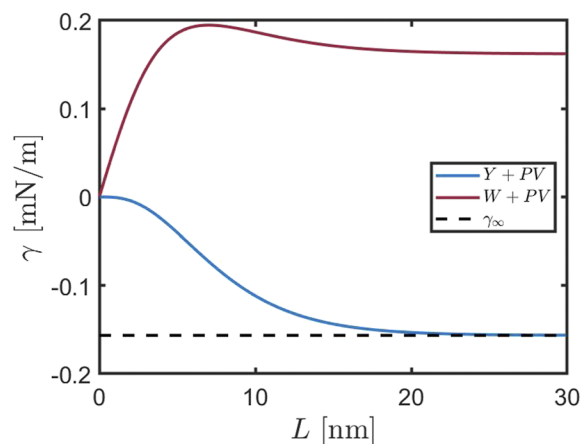


FIG. 5. Surface tension of a confined electrolyte solution for varying plate separation calculated using Eq. (43) (blue) and $\gamma A = W + PV$ (red). The surface tension at infinite separation γ_{∞} is calculated using Eq. (47). The surfaces have fixed surface potential $\mathcal{V} = 1 \text{ kT}/e$. The solution conditions are $c_B = 0.01 \text{ M}$, $T = 300 \text{ K}$, and $l_B = 0.7 \text{ nm}$.

the range of validity of the PB theory). At low surface charge density, the surface tension increases in magnitude with decreasing bulk concentration because the surface potential is linearly increasing in the Debye length. The crossover from the quadratic to the linear regimes occurs at a lower surface charge density for low concentrations.

Now, we move to the case of the surface tension at a finite separation length between the plates. We examine its variation with the plate separation; the results are shown in Fig. 5, where the blue line is the correct calculation of the surface tension by Eq. (43), the red line is calculated incorrectly from the grand potential $\gamma A = W + PV$, and the black dotted line is the value of the surface tension at infinite separation. Clearly, the expression using Y captures the predicted behavior for large L by Eq. (47). For the particular conditions in Fig. 5, using W instead of Y would lead to qualitatively different conclusions with regard to both the sign and the trend. We highlight these results for using the wrong expression $\gamma A = W + PV$ to heed caution in calculating the surface tension. As discussed in the Introduction, the literature is not always clear in what is meant by “grand potential,” often using the same language for W and Y -type potentials; only the latter is the correct potential for computing the interfacial tension.

We end this section by examining mixed boundary conditions—the left plate at constant surface charge density and the right plate at constant surface potential. Taking advantage of the framework developed earlier, we obtain the corresponding thermodynamic potential by Legendre transforming W for the right plate with specified surface potential. The free energy and pressure are thus given by

$$X = W - e\mathcal{V}_2 Q_2, \quad (52)$$

$$PA = - \left(\frac{\partial X}{\partial L} \right)_{T, A, \mu_{\pm}, Q_1, V_2} \quad (53)$$

Figure 6(a) shows the disjoining pressure as a function of plate separation with mixed boundary conditions. From Fig. 6(a), we

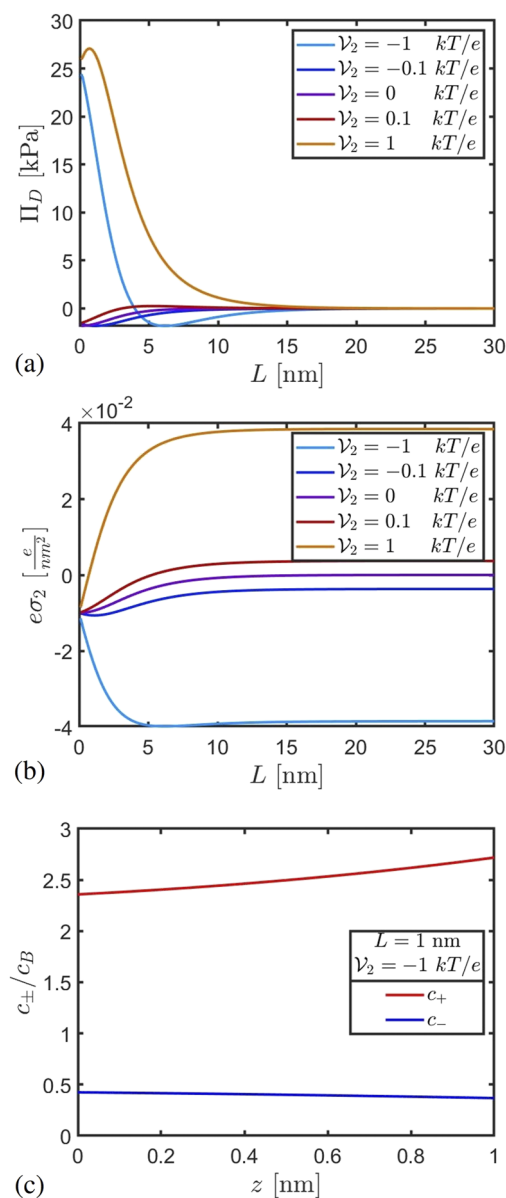


FIG. 6. Disjoining pressure and surface charge density for an electrolyte solution with the left plate at constant surface charge density $e\sigma_1 = 0.01 e/nm^2$ and the right plate at various constant surface potentials. The solution conditions are $c_B = 0.01$ M, $T = 300$ K, and $l_B = 0.7$ nm. (a) The disjoining pressure was calculated by numerically differentiating X given by Eq. (52). (b) Surface charge density on the right plate vs plate separation L . (c) Positive and negative ion density profiles vs distance from the plate at separation $L = 1$ nm and right plate surface potential $V_2 = -1 kT/e$.

see that the profiles saturate at zero separation. As discussed earlier, this is best interpreted in terms of the solvation energy of two plates, where one plate is at fixed surface charge density and the other is at fixed surface potential. Furthermore, from Fig. 6(b), the surface charge density on the right plate at zero separation is simply equal and opposite to the surface charge density of the left plate to satisfy charge neutrality. From Fig. 6, the most intriguing result is the pressure profile for $V_2 = -1 kT/e$. At moderate separation, the constant surface potential plate is negatively charged so the plates are attractive as expected. However, at close separation (<4 nm), the plates become repulsive despite still being oppositely charged as shown in Fig. 6(b). Surprisingly, the ion density profiles at separation $L = 1$ nm in Fig. 6(c) reveal that the ion layer present on the surface of the left plate undergoes charge inversion—that is, the ion layer switches from accumulation of negative ions to accumulation of positive ions on the positively charged left plate, causing “unlike-charge repulsion” at close separation. This phenomenon at the Poisson–Boltzmann level was first reported in Refs. 42 and 79. It is important to emphasize that, while unlike-charged plates can repel, like-charged plates can never attract at the Poisson–Boltzmann level, although like-charged attraction is possible at the Poisson–Boltzmann level in curved geometries.⁸⁰

VIII. CONCLUSION

In this work, we present a pedagogical framework to clarify the subtleties of specified surface charge and specified surface potential conditions using a variational approach and the Legendre transform. For specified surface charge, both the Poisson–Boltzmann equation and fixed surface charge boundary conditions follow naturally from variation of the grand free energy W . For specified surface potential, we demonstrate that the Poisson–Boltzmann equation and constant surface potential boundary conditions are natural consequences of variation of the thermodynamic potential Y —a new thermodynamic potential obtained from the Legendre transform of W . We further verify the thermodynamic consistency of our approach by showing both W and Y result in the same pressure equation of state. The thermodynamic potential Y is shown to be the natural connection to interfacial tension, regardless of whether the surface is characterized by constant surface charge or constant surface potential. Finally, we provide some illustrative examples to emphasize the importance of properly treating the free energy.

The framework presented can be directly applied to any electrostatic system in parallel plate geometry, even with electrostatic correlations, such as image charge. For example, it can be applied to calculating work in charging–discharging processes for capacitance application, as well as phase equilibrium in electrowetting of surfaces.^{81,82} In the electrowetting of surfaces, an electrolyte solution in contact with a surface at fixed surface potential would require a Y like free energy for determining the transition. In its current form, our framework is limited to slab geometries, as the system size is characterized conveniently by area A and separation L . It will be interesting to extend the framework to curved geometries, where Ref. 80 predicts like-charge attraction between two metal spheres even at the Poisson–Boltzmann level. We reserve such a task for future work.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

D.B. and C.B. contributed equally to this work.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: DERIVATION OF THE INTERPLAY PRESSURE

Here, we present a derivation of Eq. (38) by application of the derivative from Eq. (37) to Y , given by Eq. (35). Although the derivation has been done before,^{28,83} we provide it here for completeness. We start with the Poisson–Boltzmann equation, given by

$$\frac{d^2\psi}{dz^2} = 8\pi l_B c_B \sinh(\psi). \quad (\text{A1})$$

Rewriting the left-hand side as $\frac{d^2\psi}{dz^2} = \frac{1}{2} \frac{d}{d\psi} \left(\frac{d\psi}{dz} \right)^2$ and integrating gives

$$-\frac{1}{8\pi l_B} \left(\frac{d\psi}{dz} \right)^2 + 2c_B \cosh(\psi) = \text{constant}. \quad (\text{A2})$$

The constant can be found by evaluating Eq. (A2) at any point in the confined system. We evaluate it at some arbitrary position $z = \xi$ between the two plates and denote its value as $C(\xi)$; thus,

$$C(\xi) = -\frac{1}{8\pi l_B} \left(\frac{d\psi}{dz} \right)^2 + 2c_B \cosh(\psi). \quad (\text{A3})$$

It is important to stress that although $C(\xi)$ is constant in position z , it still depends on plate separation L .

Using Eq. (A3) in Eq. (35), we obtain

$$\beta Y/A = -LC(\xi) - \frac{1}{4\pi l_B} \int_0^L dz \left(\frac{d\psi}{dz} \right)^2. \quad (\text{A4})$$

Next, we rewrite the last term of Eq. (A4) as an integral over ψ instead of z by making use of Eq. (A3),

$$\beta Y/A = -LC(\xi) - \frac{1}{4\pi l_B} \int_{\psi_0=\psi_1}^{\psi_L=\psi_2} d\psi \sqrt{16\pi l_B c_B \cosh(\psi) - 8\pi l_B C(\xi)}. \quad (\text{A5})$$

Applying the derivative from Eq. (37) to Eq. (A5) yields

$$\beta P = - \left(\frac{\partial \beta Y/A}{\partial L} \right)_{T, A, \mu_{\pm}, \psi_1, \psi_2} = C(\xi) + L \frac{\partial C(\xi)}{\partial L} - \frac{\partial C(\xi)}{\partial L} \int_{\psi_1}^{\psi_2} d\psi \frac{1}{\sqrt{16\pi l_B c_B \cosh(\psi) - 8\pi l_B C(\xi)}}. \quad (\text{A6})$$

Upon switching the integration in Eq. (A6) back to z , the integral yields simply L , resulting in exact cancellation of the last two terms. Therefore,

$$\beta P = C(\xi) = -\frac{1}{8\pi l_B} \left(\frac{d\psi}{dz} \right)^2 + 2c_B \cosh(\psi). \quad (\text{A7})$$

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